4.4.2 Electromagnetic waves

	Learning outcomes	Additional guidance
	Learners should be able to demonstrate and apply their knowledge and understanding of:	
(a)	electromagnetic spectrum; properties of electromagnetic waves	
(b)	orders of magnitude of wavelengths of the principal radiations from radio waves to gamma rays	
(c)	plane polarised waves; polarisation of electromagnetic waves	Learners will be expected to know about polarising filters for light and metal grilles for microwaves in demonstrating polarisation.
(d)	**(i)** refraction of light; refractive index; $n = \frac{c}{v}$; $n \sin \theta$ = constant at a boundary where θ is the angle to the normal **(ii)** techniques and procedures used to investigate refraction and total internal reflection of light using ray boxes, including transparent rectangular and semi-circular blocks	
(e)	critical angle; $\sin C = \frac{1}{n}$; total internal reflection for light.	